TONIC: a structure-preserving operator for families of optimal control problems

Concept map of the framework and the main empirical findings

Problem family

Parameterized multi-agent optimal control

- initial state
- obstacle geometry
- agent capability
- drag / dynamics

Tasks

- collision avoidance
- obstacle avoidance
- target reaching

family of instances $\theta \in \Theta$

PMP solution operator

Optimal control problem

state costs $F_i(x_i)$
control costs $G_i(u_i)$
interaction term $W(x)$

Pontryagin maximum principle

$$\dot{x} = \nabla_p H(x,p)$$
$$\dot{p} = -\nabla_x H(x,p)$$

learn the solution operator

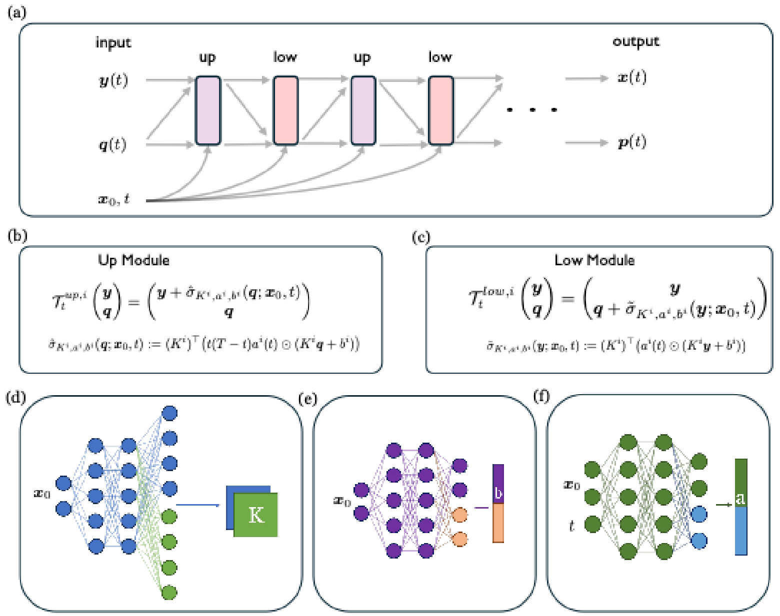
$S[\theta] : \theta \mapsto (\bar{x}\theta(t), \bar{p}\theta(t))$

TONIC

1. latent right-space solver

LQR • rotational LQR • pretrained TSympOCNet

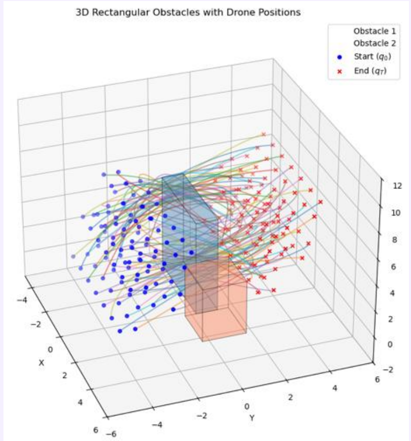
2. conditional symplectic operator



physics-informed training: minimize PMP residuals

Key findings

- Generalizes across initial conditions, geometry, and heterogeneity
- Near-optimal cost with high feasibility
- **21–4600× faster online than baselines**



results span free space, obstacle avoidance, maze navigation, varying geometry, and heterogeneous swarms

Operator view: $\theta \in \Theta$ ➤ $(\bar{y}\theta(t), \bar{q}\theta(t))$ ➤ $(\bar{x}\theta(t), \bar{p}\theta(t))$ *learn one operator, reuse it across a family of instances*